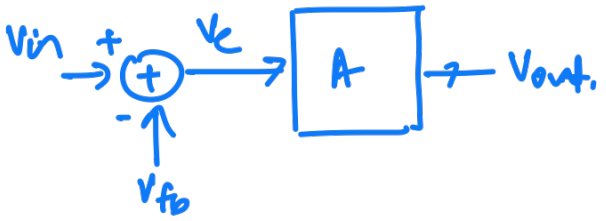
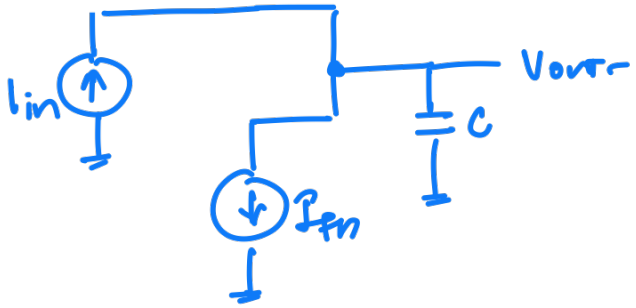
## Lecture # 25




$V_{out} = A(V_{in} - V_{fb})$

if $A \to \infty$,




$$V_{out}(t) = \frac{1}{C}\int_0^t (I_{in} - I_{fb})\,dt$$

At steady state,

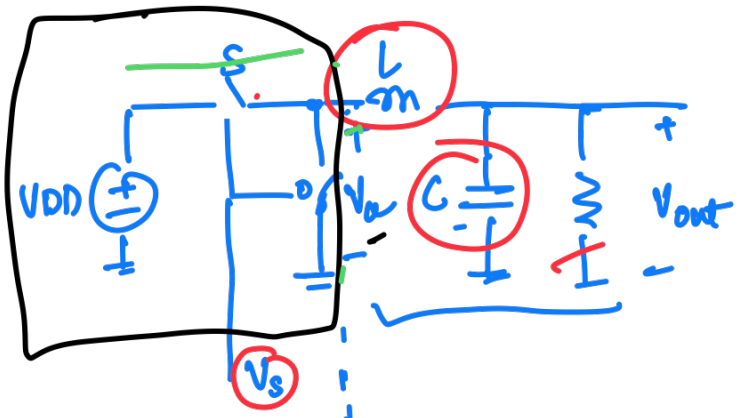
$V_{out}$ (of non-zero value

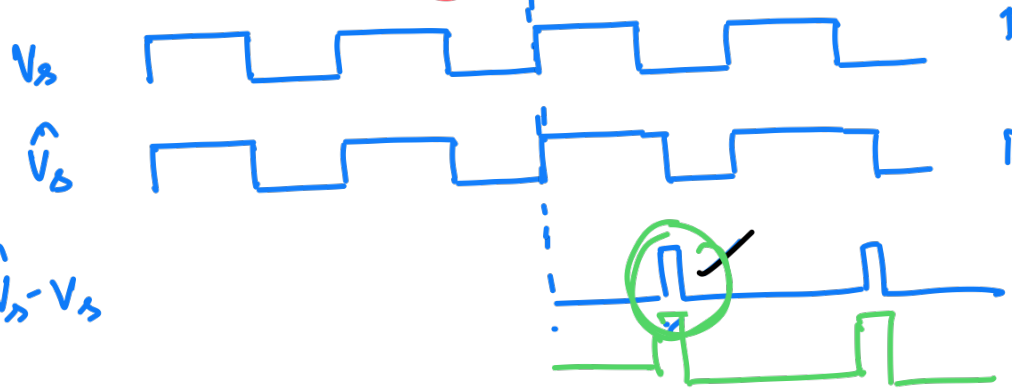
if $I_{in} - I_{fb} = 0$

$t \to \infty$

$$V_{out}(s) = (I_{in} - I_{fb})\frac{1}{sC} = \frac{1}{s}\left(\frac{I_{in} - I_{fb}}{C}\right)$$

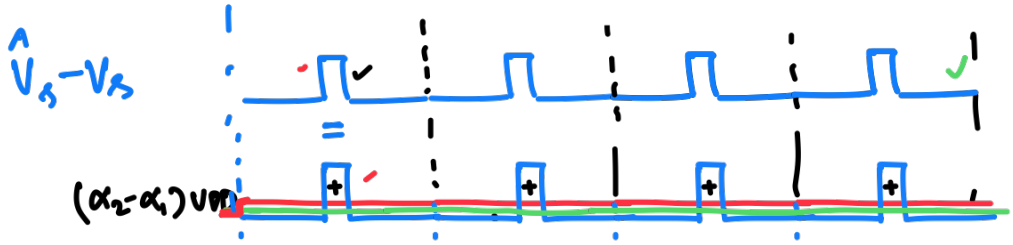
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$$\Delta V_{out}(t) = (\hat{V}_s - V_s)(t) * h(t)$$

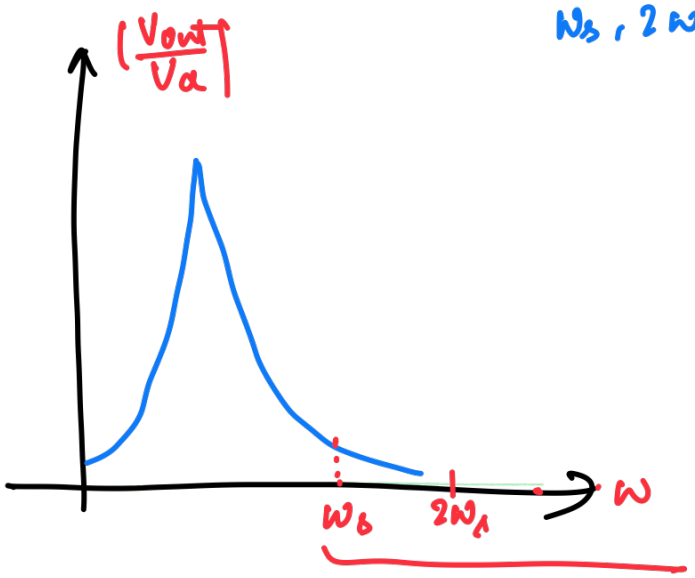
$$= \left\{ \sum a_n \cos(n\omega_s t) + b_n \sin(n\omega_s t) \right\} * h(t)$$

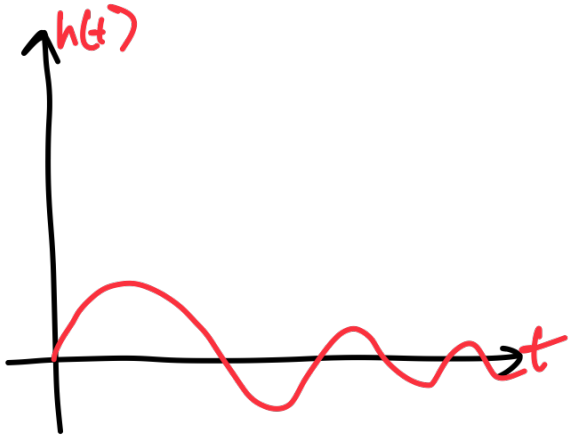



$$\Delta V_{out}(t) = (\alpha_2 - \alpha_1) V_{DD}\, u(t) * h(t) +$$

$$\left\{ \sum a_n \cos(n\omega_s t) + b_n \sin(n\omega_s t) \right\} * h(t)$$

$\omega_s, 2\omega_s, \ldots$

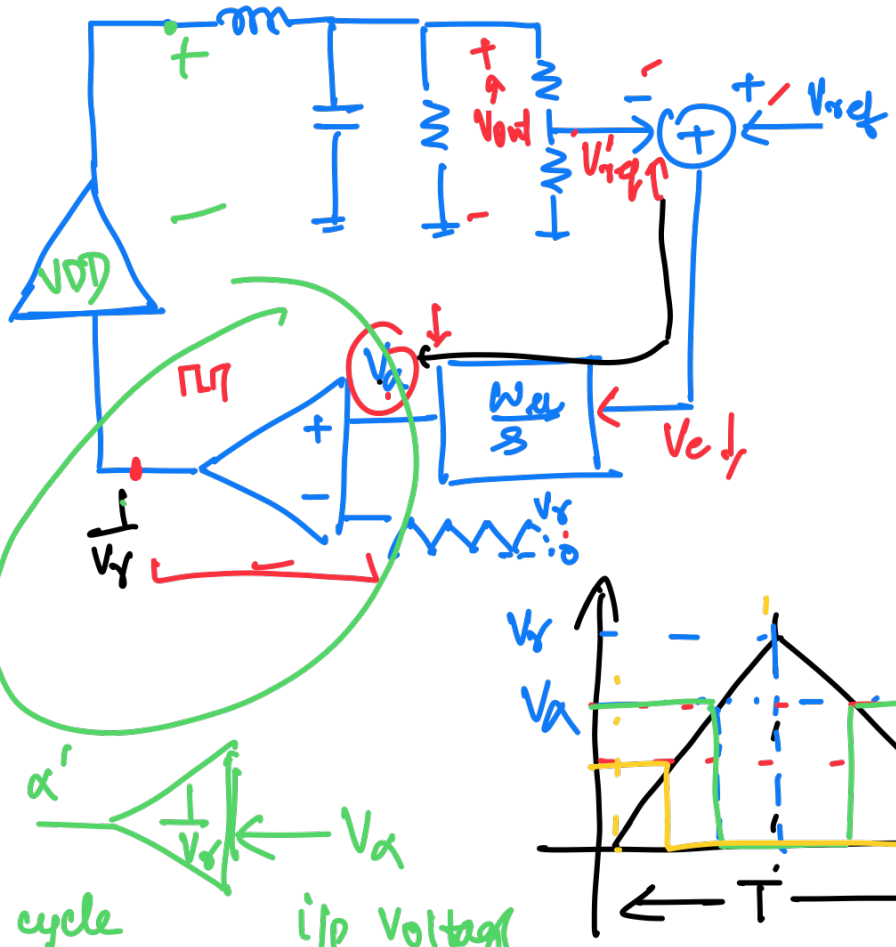


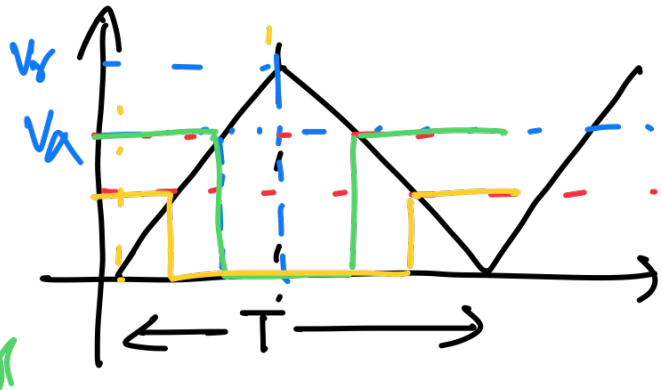




$$\frac{V_{out}}{V_a} = \frac{\left(\frac{1}{sC} \| R\right)}{sL + \frac{1}{sC} \| R} = \frac{1}{s^2 LC + \frac{sL}{R} + 1}$$

$$= \frac{R/(1+sCR)}{\frac{R}{sCR+1} + sL}$$

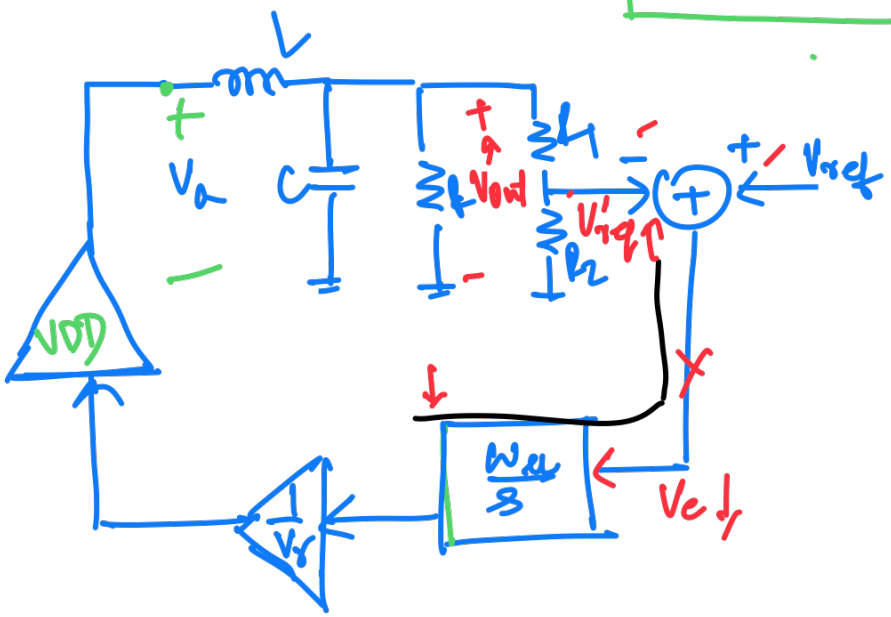
$$= \frac{R}{s^2 LCR + sL + R}$$



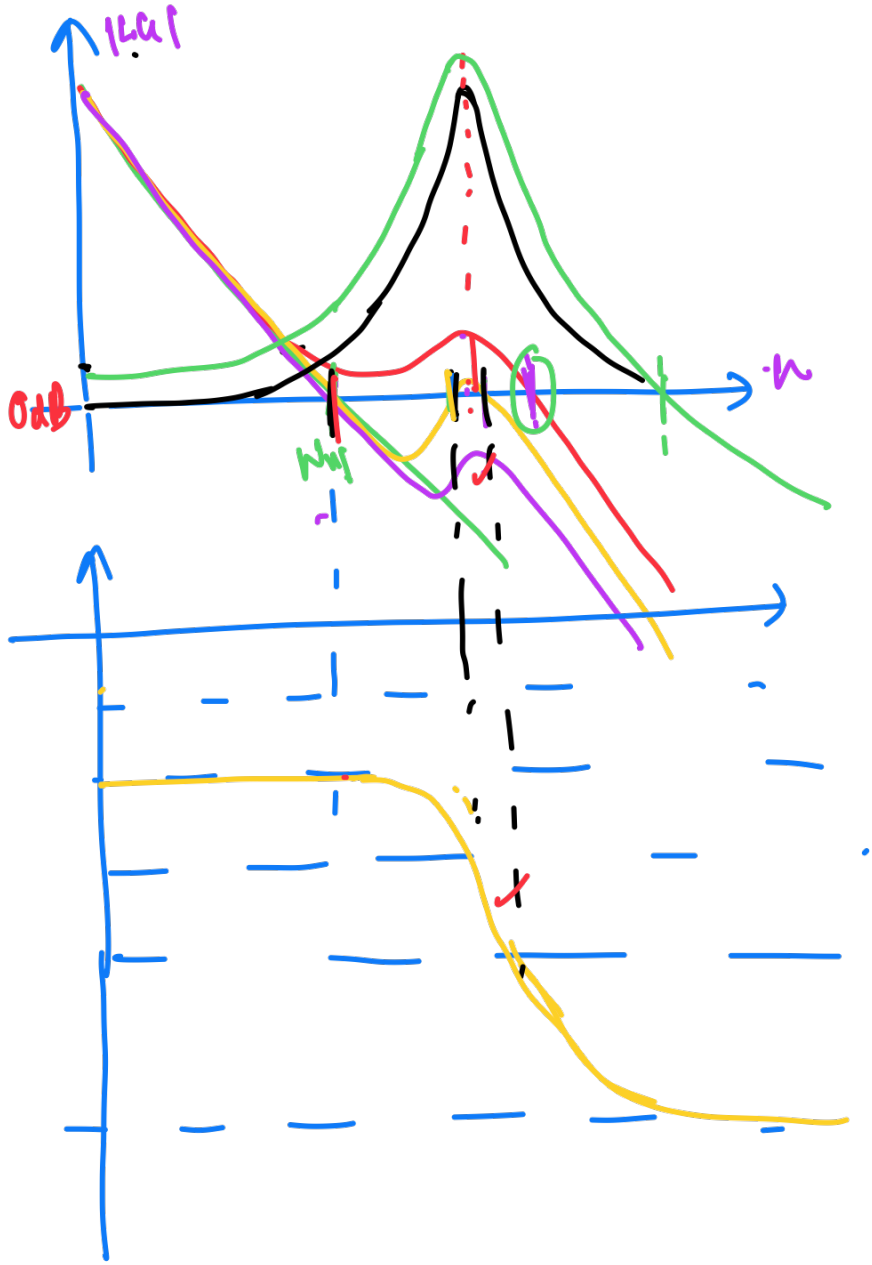

$$\frac{V_r}{T/2} = \frac{V_\alpha}{T_H/2}$$

$$\frac{T_H}{T} = \frac{V_\alpha}{V_r}$$




$$\frac{R_2}{R_1+R_2} = K$$

$$LG = \frac{\omega_u}{s} \times \frac{1}{V_r} \times VDD \times \frac{1}{s^2LC + \frac{sL}{R} + 1} \times \frac{1}{K}$$




$$\frac{V_{DD}}{V_\gamma} \times \frac{1}{K} \times \frac{W_4}{L_4} = H_1$$

---

$$H_2(s) = \frac{1}{1 + \frac{s}{\omega_p Q_p} + \frac{s^2}{\omega_p^2}}$$

$$|H(j\omega)|^2 = \left(1 - \frac{\omega^2}{\omega_p^2}\right)^2 + \left(\frac{\omega}{\omega_p}\right)^2 \times \frac{1}{Q_p^2}$$

$$\frac{d(\quad)}{d\omega} = 0$$

- $\omega_{max}$, $|H_2|_{max}$

$$|H_1(\omega_{max})| < |$$

$$Q_p \, |H_1(\omega_{max})| < |$$